

HumanoidGym-Ex README

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HumanoidGym-Ex

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HumanoidGym-Ex is not a new humanoid learning framework from scratch. It is a Humanoid-Gym-style extension framework that preserves the original Humanoid-Gym user experience while enabling future IsaacLab and Genesis backends.

The main work in this repository is the migration of the original Humanoid-Gym codebase plus a Humanoid-Gym-style IsaacLab Direct backend. IsaacGym remains the compatibility baseline; IsaacLab / Isaac Sim is the primary extension path.

HumanoidGym-Ex 不是一个从零重写的人形机器人强化学习框架。它是一个 **Humanoid-Gym 风格的扩展框架**：尽量保留原版 Humanoid-Gym 的脚本入口、配置方式、reward 写法、PPO 接口和使用习惯，同时逐步支持 IsaacGym、IsaacLab Direct Workflow，并为 Genesis 后端预留接口。

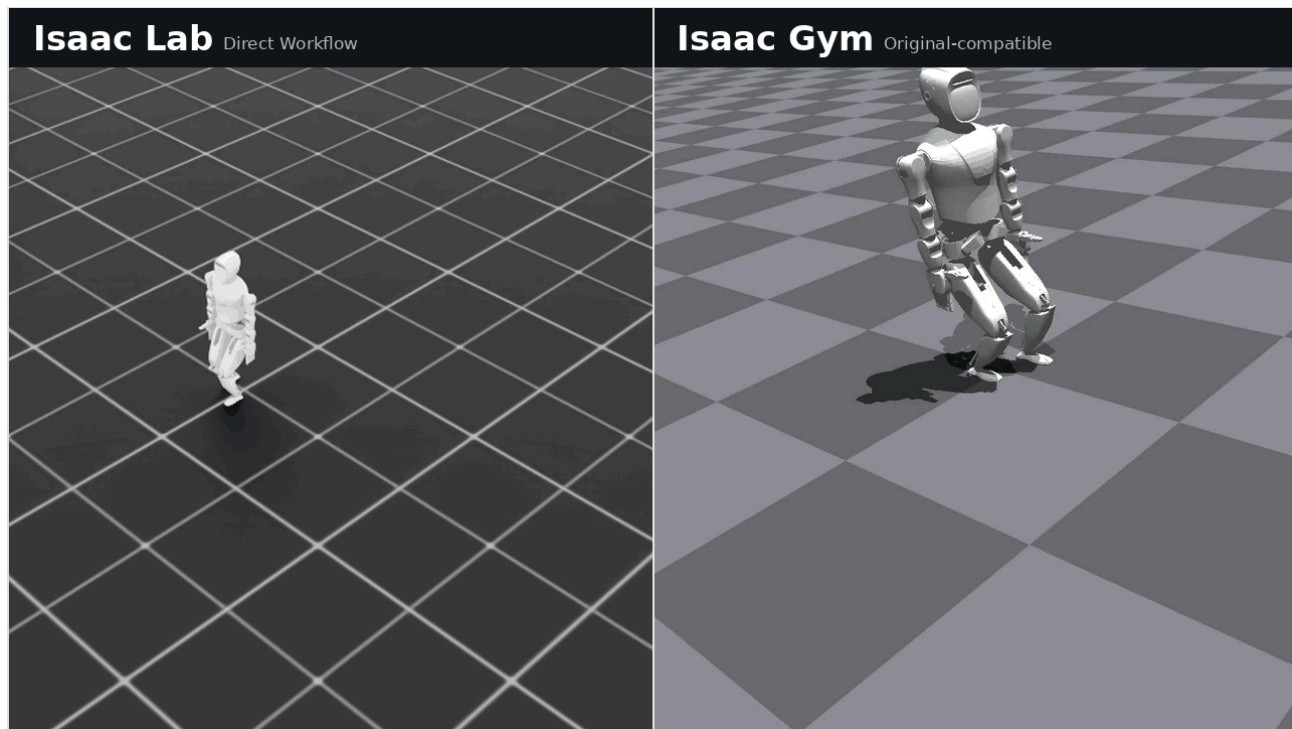
本仓库的主线工作是：迁移原版 Humanoid-Gym，并在不破坏原使用体验的前提下新增 IsaacLab Direct 支持。IsaacGym 是兼容基线，IsaacLab / Isaac Sim 是主要扩展方向。

本项目由 **灯哥开源** 移植与维护。

- B 站主页：<https://space.bilibili.com/493192058>
- 迁移目标：让熟悉 Humanoid-Gym 的用户可以用接近原版的方式进入 IsaacLab / Isaac Sim 生态。

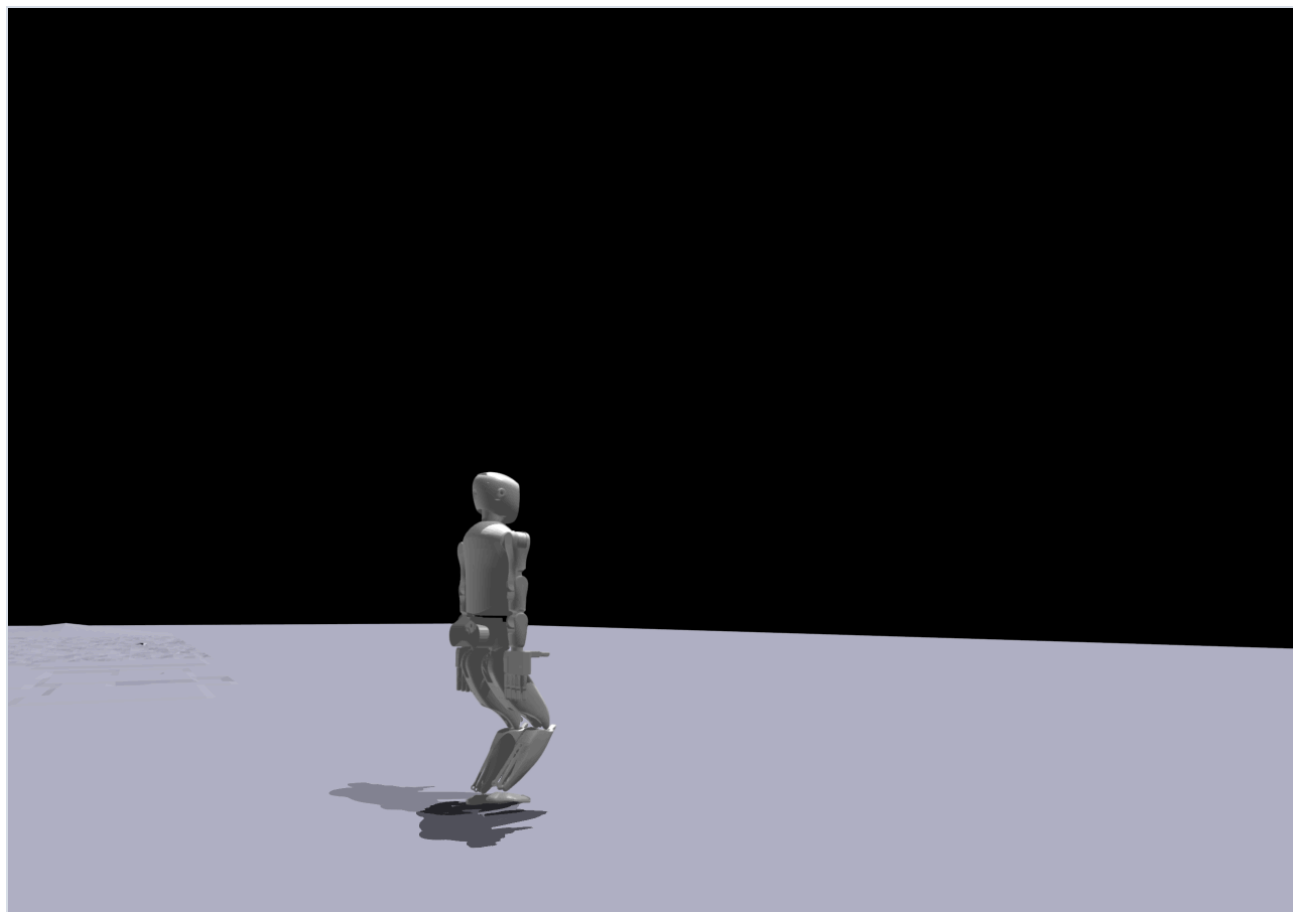
项目截图

IsaacLab Direct 与 IsaacGym 原版兼容路径的训练后策略并列播放：

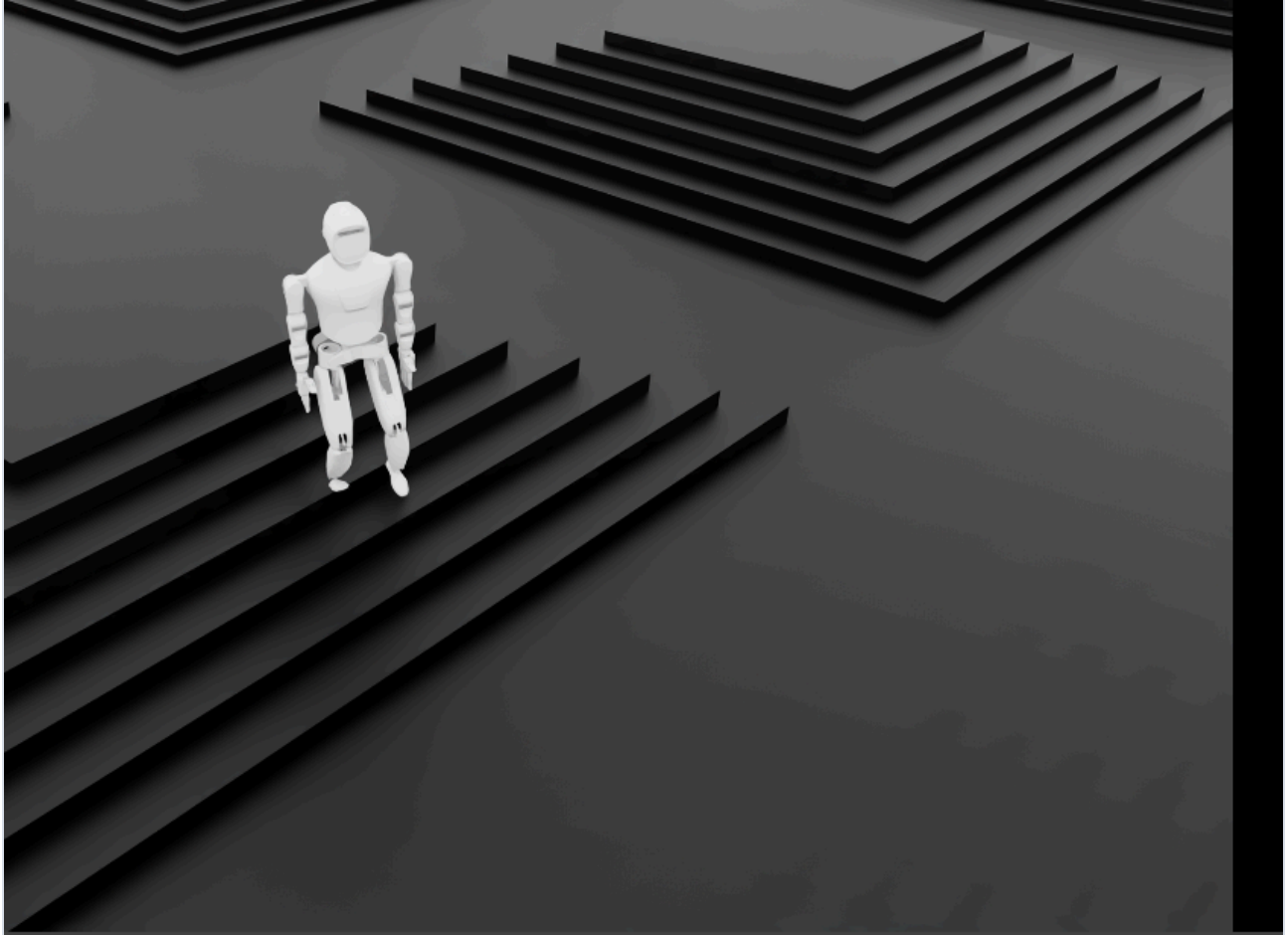


静态截图：

IsaacGym 训练后策略播放：



IsaacLab / Isaac Sim 训练后策略播放：



为什么做 HumanoidGym-Ex

原版 Humanoid-Gym 的优点是非常直接：

- `train.py / play.py` 脚本中心化，学习成本低。
- `robot config`、`reward scale`、`observation`、`reset`、`command curriculum` 都在熟悉的位置。
- `reward` 函数集中在环境类里，便于调试和快速实验。
- PPO 接口基于 `rs1_rl` 风格，训练链路简单。

IsaacLab / Isaac Sim 的生态更现代，但 Manager-based workflow 对很多从 Humanoid-Gym / Legged-Gym 迁移过来的用户来说会显得拆分较多。HumanoidGym-Ex 的目标不是把 Humanoid-Gym 改造成一个全新的大框架，而是在保留原有使用体验的前提下，逐步接入新后端。

当前状态

模块	状态
IsaacGym 后端	已可训练、play、导出策略
IsaacLab Direct 后端	已可 smoke、训练、play、导出策略
Genesis 后端	已预留接口，暂未完整实现
XBot-L 原版示例	已迁移
MuJoCo sim2sim	已迁移原版脚本
PPO	保留本地 Humanoid-Gym / rsl_rl 风格接口
reward parity	默认 reward 名称和默认 reward scale 保持一致
rough terrain	Gym / Lab 均可运行，长训练曲线仍有差距
measured heights	Gym / Lab 均支持，actor 705，critic 780

默认任务名：

```
humanoid_ppo
```

默认示例机器人：

```
RobotEra XBot-L
```

设计原则

HumanoidGym-Ex 的核心原则是：

- 1. **不重写用户习惯**
熟悉 Humanoid-Gym 的用户应该能快速找到 train、play、cfg、reward、obs、reset、commands。
- 2. **不引入复杂插件系统**
backend interface 只抽象必要接口，不把项目做成过度工程化的大平台。
- 3. **不把 reward / obs 拆成 Manager-based 风格**
reward 仍然集中、直观，便于从原版代码迁移。

4. IsaacLab 使用 Direct Workflow

IsaacLab 路径基于 DirectRLEnv、Articulation、Scene、ContactSensor 和 RayCaster，不使用 Manager-based task。

5. 默认 reward 保持一致

IsaacGym 和 IsaacLab 默认使用同一套 reward 名称和 reward scale。当前调试优先做数据顺序、接触、地形、reset、随机化和物理参数对齐，而不是为某个后端单独改 reward。

目录结构

```
humanoid_gym_ex/  
├─ humanoid_gym_ex/  
│   ├── algo/                # PP0 / actor-critic / runner  
│   ├── envs/  
│   │   ├── base/            # Humanoid-Gym style base env/config/backend interface  
│   │   ├── backends/        # IsaacGym / IsaacLab / Genesis adapters  
│   │   └─ robots/           # XBot-L and future humanoids  
│   ├── resources/           # URDF / meshes / MuJoCo resources  
│   ├── scripts/             # train/play/sim2sim/diagnostics  
│   └─ utils/  
├─ docs/  
├─ images/  
├─ logs/  
└─ README.md
```

安装

IsaacGym 环境

假设你已经安装 IsaacGym Preview 4，并有一个类似 legged_gym 的 conda 环境：

```
conda activate legged_gym  
pip install -e .
```

IsaacLab / Isaac Sim 环境

IsaacLab 建议使用单独 conda 环境。本仓库测试时使用：

```
/home/cra02/anaconda3/envs/env_isaaclab
```

运行 IsaacLab 脚本时建议显式设置 PYTHONPATH:

```
export PYTHONPATH=/path/to/HumanoidGym-Ex
```

IsaacLab Direct 使用方式

IsaacLab 是 HumanoidGym-Ex 在原版 Humanoid-Gym 迁移基础上新增的核心能力。本路径不使用 Manager-based task, 而是保留 Humanoid-Gym 风格的上层 env / reward / obs / reset 写法。

原版 IsaacGym 训练 / play 启动方式仍然保留, 但已放入折叠区: [IsaacGym 原版兼容训练 / play 命令](#)。

IsaacLab PPO smoke

```
PYTHONPATH=/path/to/HumanoidGym-Ex \  
WANDB_MODE=offline \  
conda run -p /path/to/env_isaaclab \  
python humanoid_gym_ex/scripts/train_isaaclab.py \  
  --headless \  
  --num_envs 64 \  
  --num_steps_per_env 60 \  
  --max_iterations 1 \  
  --run_name isaacrab_ppo_smoke
```

IsaacLab 训练

```
PYTHONPATH=/path/to/HumanoidGym-Ex \  
WANDB_MODE=offline \  
conda run -p /path/to/env_isaaclab \  
python humanoid_gym_ex/scripts/train_isaaclab.py \  
  --headless \  
  --num_envs 4096 \  
  --num_steps_per_env 60 \  
  --max_iterations 1000 \  
  --seed 42 \  
  --run_name xbot_isaacrab_1000 \  
  --device cuda:0
```

IsaacLab rough terrain + measured heights

```
PYTHONPATH=/path/to/HumanoidGym-Ex \  
WANDB_MODE=offline \  
conda run -p /path/to/env_isaaclab \  
python humanoid_gym_ex/scripts/train_isaaclab.py \  
  --headless \  
  --num_envs 4096 \  
  --num_steps_per_env 60 \  
  --max_iterations 1000 \  
  --seed 42 \  
  --run_name rough_heights_curric_lab \  
  --terrain rough \  
  --measure_heights \  
  --terrain_curriculum \  
  --device cuda:0
```

rough measured-height 模式下:

```
actor policy obs: 705  
critic obs:      780
```

这与原版 Humanoid-Gym XBot 路径一致: actor 不拼接 height samples, critic privileged obs 拼接 height samples。

IsaacLab Play

```
PYTHONPATH=/path/to/HumanoidGym-Ex \  
conda run -p /path/to/env_isaaclab \  
python humanoid_gym_ex/scripts/play_isaaclab.py \  
  --load_run <run_dir_name> \  
  --checkpoint 1000 \  
  --fix_command \  
  --follow_camera \  
  --terrain rough \  
  --measure_heights \  
  --terrain_curriculum \  
  --device cuda:0
```

IsaacLab viewer 默认开启近距离可缩放跟随相机:

- 默认 --camera_zoom 0.45
- 鼠标滚轮可以继续自由缩放
- 相机会跟随机器人 root, 但不会强制覆盖你手动调整后的 zoom

► IsaacGym 原版兼容训练 / play 命令 (点击展开)

与原版 Humanoid-Gym 的兼容目标

HumanoidGym-Ex 尽量保持：

- 原版 task registry 风格
- 原版 nested Python cfg 风格
- 原版 `cfg.rewards.scales` reward dispatch
- 原版 observation / privileged observation 维度
- 原版 PPO runner 使用方式
- 原版 XBot-L 资源路径和 MuJoCo sim2sim 示例
- 原版脚本中心化体验

默认 XBot-L 维度：

```
num_actions = 12
num_single_obs = 47
frame_stack = 15
num_observations = 705
single_num_privileged_obs = 73
c_frame_stack = 3
num_privileged_obs = 219
rough measured-height critic obs = 780
```

后端抽象

当前 backend interface 保持极简：

```
create_sim()
create_envs()
step(actions)
reset(env_ids)
get_root_states()
get_dof_pos()
get_dof_vel()
get_contact_forces()
set_dof_targets()
apply_domain_randomization()
render_or_viewer_step()
```

设计目的不是构建复杂插件系统，而是让上层 Humanoid-Gym 风格环境尽量不感知 IsaacGym / IsaacLab 的底层差异。

文档

- [Architecture review](#)
- [Backend interface](#)
- [Design goals](#)
- [IsaacLab migration notes](#)
- [Roadmap](#)

Roadmap

- Phase 0: 原版 Humanoid-Gym 代码迁移。
- Phase 1: IsaacGym backend adapter，保持原版训练和 play 可运行。
- Phase 2: IsaacLab Direct Workflow backend，支持 XBot smoke / train / play / rough terrain。
- Phase 3: Genesis backend 接口预留。
- Phase 4: 更多机器人、更多原版场景和跨后端长训练统计。

致谢与声明

本项目基于原版 `roboterax/humanoid-gym` 进行迁移和扩展，并保留迁移源文件中的 BSD-3-Clause license headers。

特别感谢：

- Humanoid-Gym 原作者与 RobotEra XBot-L 示例。
- Legged-Gym / rsl_rl 生态。
- NVIDIA IsaacGym、IsaacLab、Isaac Sim。
- 灯哥开源社区。

灯哥开源 B 站视频号：

<https://space.bilibili.com/493192058>

HumanoidGym-Ex English README

HumanoidGym-Ex is not a new humanoid learning framework from scratch. It is a Humanoid-Gym-style extension framework that preserves the original Humanoid-Gym user experience while enabling future IsaacLab and Genesis backends.

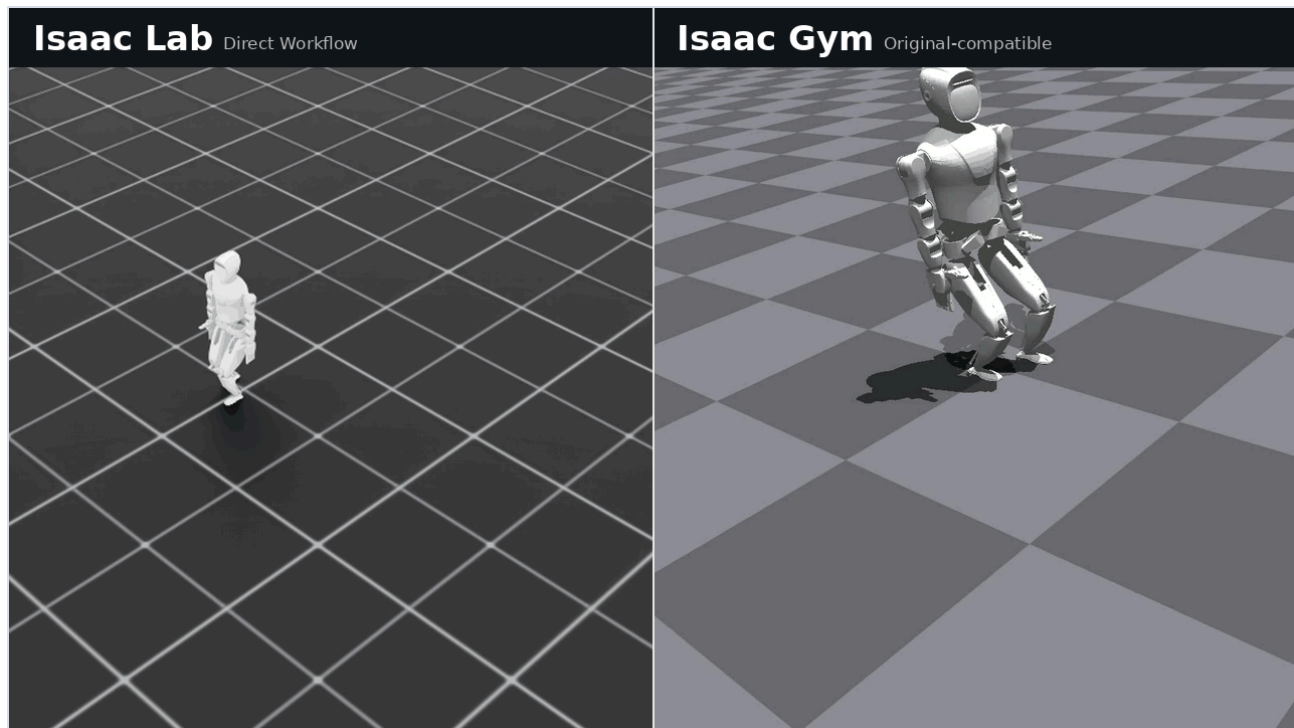
The main work in this repository is the migration of the original Humanoid-Gym codebase plus a Humanoid-Gym-style IsaacLab Direct backend. IsaacGym remains the compatibility baseline; IsaacLab / Isaac Sim is the primary extension path.

This project is an open-source migration by **Deng Ge Open Source**.

- Bilibili: <https://space.bilibili.com/493192058>
- Goal: keep the original Humanoid-Gym workflow familiar while making IsaacLab / Isaac Sim migration practical.

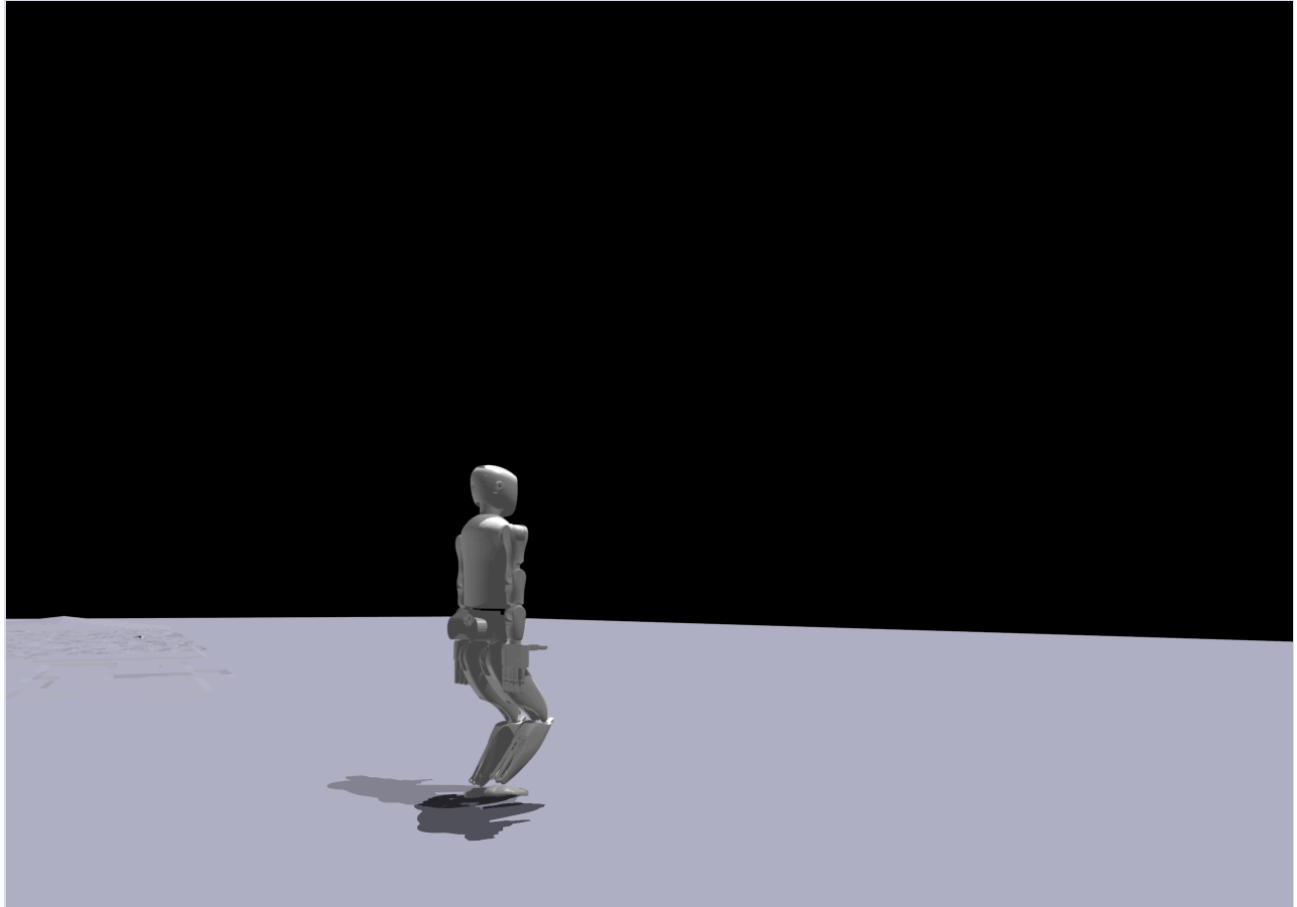
Screenshots

Side-by-side trained policy playback in IsaacLab Direct and the IsaacGym-compatible path:

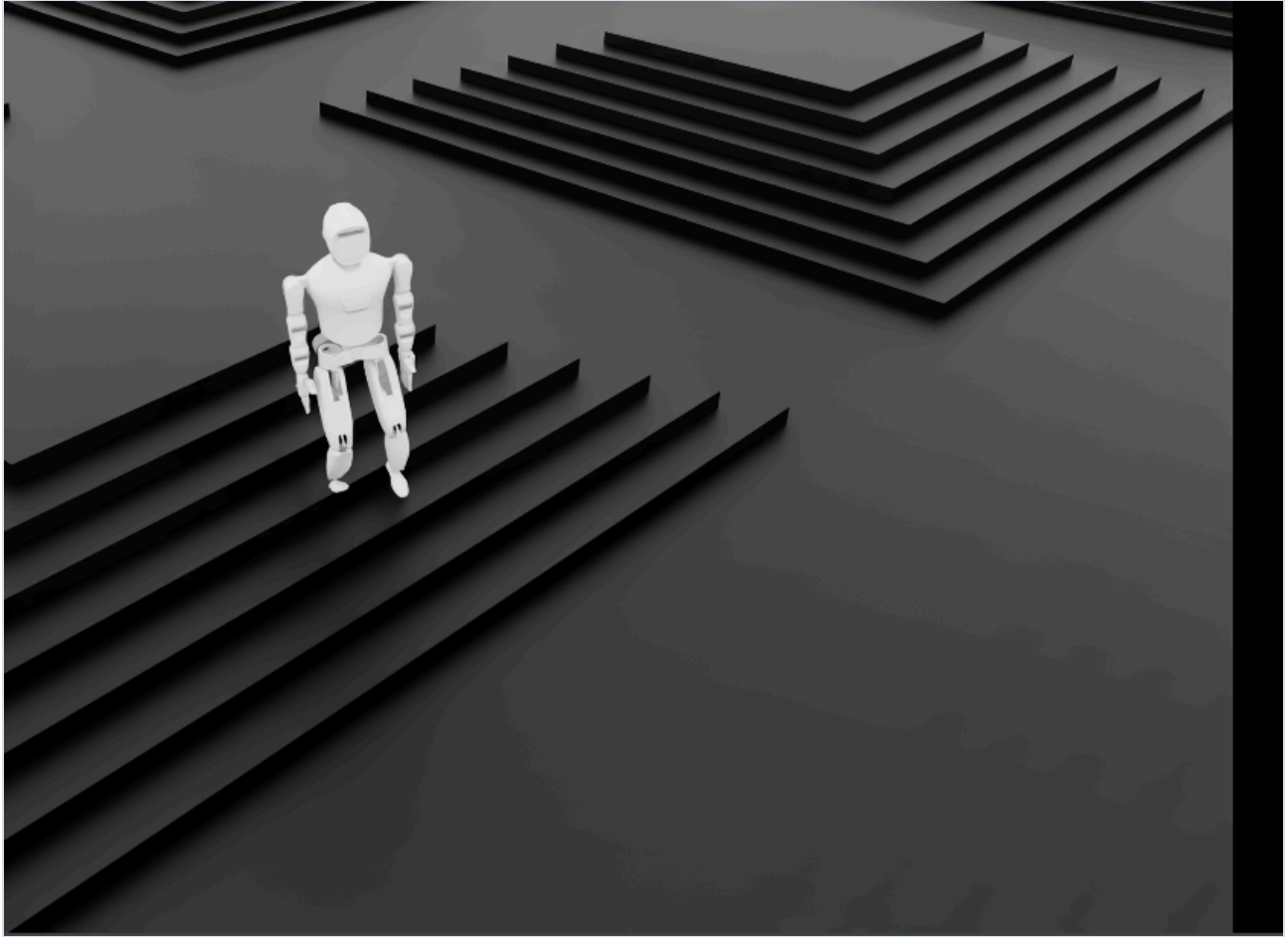


Static screenshots:

Trained policy play in IsaacGym:



Trained policy play in IsaacLab / Isaac Sim:



Motivation

The original Humanoid-Gym is practical because it is script-centered and easy to modify:

- `train.py` and `play.py` are the main user entry points.
- Robot config, reward scales, observations, resets, and commands are easy to find.
- Reward functions are written in one environment class instead of being split across many manager terms.
- The PPO runner follows the familiar `rs_l_rl` style.

HumanoidGym-Ex keeps that style. It does not try to replace Humanoid-Gym with a new framework. Instead, it adds a conservative backend path toward IsaacLab Direct Workflow and, later, Genesis.

Current Status

Component	Status
IsaacGym backend	Train, play, and policy export are active
IsaacLab Direct backend	Smoke, train, play, export, rough terrain are active
Genesis backend	Interface reserved, not fully implemented
XBot-L example	Migrated from upstream Humanoid-Gym
MuJoCo sim2sim	Migrated from upstream
PPO	Local Humanoid-Gym / rsl_rl-style runner
Reward parity	Default reward names and scales are shared
Rough terrain	Runs in both backends; long-run convergence is not fully equivalent yet
Measured heights	Actor 705, critic 780 on rough terrain

Default task:

```
humanoid_ppo
```

Default example robot:

```
RobotEra XBot-L
```

Design Principles

1. Preserve the Humanoid-Gym user experience

Users should still recognize train, play, config, reward, observation, reset, and command code.

2. Keep backend abstraction small

The project intentionally avoids a complex plugin system.

3. Keep reward and observation code centralized

The IsaacLab path does not split rewards into Manager-based terms.

4. Use IsaacLab Direct Workflow

The IsaacLab implementation uses `DirectRLEnv`, `Articulation`, `Scene`, `ContactSensor`, and `RayCaster`, not Manager-based tasks.

5. Keep default rewards identical

IsaacGym and IsaacLab use the same default reward names and reward scales. Current alignment work focuses on state ordering, contact reporting, terrain sampling, reset placement, randomization, and physics settings before any backend-specific reward shaping.

Installation

For IsaacGym:

```
conda activate legged_gym
pip install -e .
```

For IsaacLab / Isaac Sim, use your IsaacLab conda environment and set:

```
export PYTHONPATH=/path/to/HumanoidGym-Ex
```

IsaacLab Direct Quick Start

Original IsaacGym train / play commands are still preserved, but they are folded here: [Original Humanoid-Gym / IsaacGym compatible commands](#).

PPO smoke:

```
PYTHONPATH=/path/to/HumanoidGym-Ex \
WANDB_MODE=offline \
conda run -p /path/to/env_isaaclab \
python humanoid_gym_ex/scripts/train_isaaclab.py \
  --headless \
  --num_envs 64 \
  --num_steps_per_env 60 \
  --max_iterations 1 \
  --run_name isaaclab_ppo_smoke
```

Train rough terrain:

```

PYTHONPATH=/path/to/HumanoidGym-Ex \
WANDB_MODE=offline \
conda run -p /path/to/env_isaacrlab \
python humanoid_gym_ex/scripts/train_isaacrlab.py \
  --headless \
  --num_envs 4096 \
  --num_steps_per_env 60 \
  --max_iterations 1000 \
  --seed 42 \
  --run_name rough_heights_curric_lab \
  --terrain rough \
  --measure_heights \
  --terrain_curriculum \
  --device cuda:0

```

Play:

```

PYTHONPATH=/path/to/HumanoidGym-Ex \
conda run -p /path/to/env_isaacrlab \
python humanoid_gym_ex/scripts/play_isaacrlab.py \
  --load_run <run_dir_name> \
  --checkpoint 1000 \
  --fix_command \
  --follow_camera \
  --terrain rough \
  --measure_heights \
  --terrain_curriculum \
  --device cuda:0

```

The IsaacLab viewer uses a close, zoomable follow camera by default:

```
--camera_zoom 0.45
```

Mouse-wheel zoom is preserved after the viewport stabilizes.

► Original Humanoid-Gym / IsaacGym compatible commands (click to expand)

Compatibility Target

HumanoidGym-Ex aims to preserve:

- upstream task registry style
- nested Python config style

- `cfg.rewards.scales` reward dispatch
- observation and privileged-observation dimensions
- PPO runner interface
- XBot-L assets and MuJoCo sim2sim example
- script-centered user workflow

Default XBot-L dimensions:

```
num_actions = 12
num_single_obs = 47
frame_stack = 15
num_observations = 705
single_num_privileged_obs = 73
c_frame_stack = 3
num_privileged_obs = 219
rough measured-height critic obs = 780
```

Documentation

- [Architecture review](#)
- [Backend interface](#)
- [Design goals](#)
- [IsaacLab migration notes](#)
- [Roadmap](#)

Credits

This project is derived from `roboterax/humanoid-gym` and preserves BSD-3-Clause license headers in migrated source files.

Thanks to:

- the original Humanoid-Gym authors and RobotEra XBot-L example
- Legged-Gym / `rsl_rl`
- NVIDIA IsaacGym, IsaacLab, and Isaac Sim
- Deng Ge Open Source community

Deng Ge Open Source Bilibili:

<https://space.bilibili.com/493192058>